VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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LAB REPORT

On

ANALYSIS AND DESIGN OF ALGORITHMS (23CS4PCADA)

Submitted by

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)
BENGALURU-560019
February to June 2026

CERTIFICATE

B. M. S. College of Engineering,
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Department of Computer Science and Engineering

This is to certify that the Lab work entitled **"ANALYSIS AND DESIGN OF ALGORITHMS"** carried out by CHETAN BARAKKI SATISH KUMAR**(1BM24CS081)**, who is bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2025-26. The Lab report has been approved as it satisfies the academic requirements in respect of Analysis and Design of Algorithms Lab - **(23CS4PCADA)** work prescribed for the said degree.

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Course outcomes:

CO1	Analyze time complexity of algorithms using asymptotic notations.
CO2	Apply various algorithm design techniques for the given problem.
CO3	Evaluate the appropriate algorithm/design technique for solving a given problem.
CO4	Conduct practical experiments by applying appropriate algorithm design technique to solve problems.

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Lab program 1:

Write program to obtain the Topological ordering of vertices in a given digraph.

```
#include <stdio.h>

int adj[100][100];
int stack[100], visited[100] = {0};
int top = -1;
int n;

void dfs(int v)
{
   visited[v] = 1;

   for (int i = 0; i < n; i++)
   {
      if (adj[v][i] && !visited[i])
      {
         dfs(i);
      }
   }

   stack[++top] = v;
}

int main()
{
   printf("Enter number of vertices: ");
   scanf("%d", &n);

   printf("Enter adjacency matrix:\n");
   for (int i = 0; i < n; i++)
   {
      for (int j = 0; j < n; j++)
      {
         scanf("%d", &adj[i][j]);
      }
   }

   for (int i = 0; i < n; i++)
   {
      if (!visited[i])
      {
         dfs(i);
      }
   }

   printf("Topological Order: ");
   for (int i = top; i >= 0; i--)
   {
      printf("%d ", stack[i]);
   }

   return 0;
```

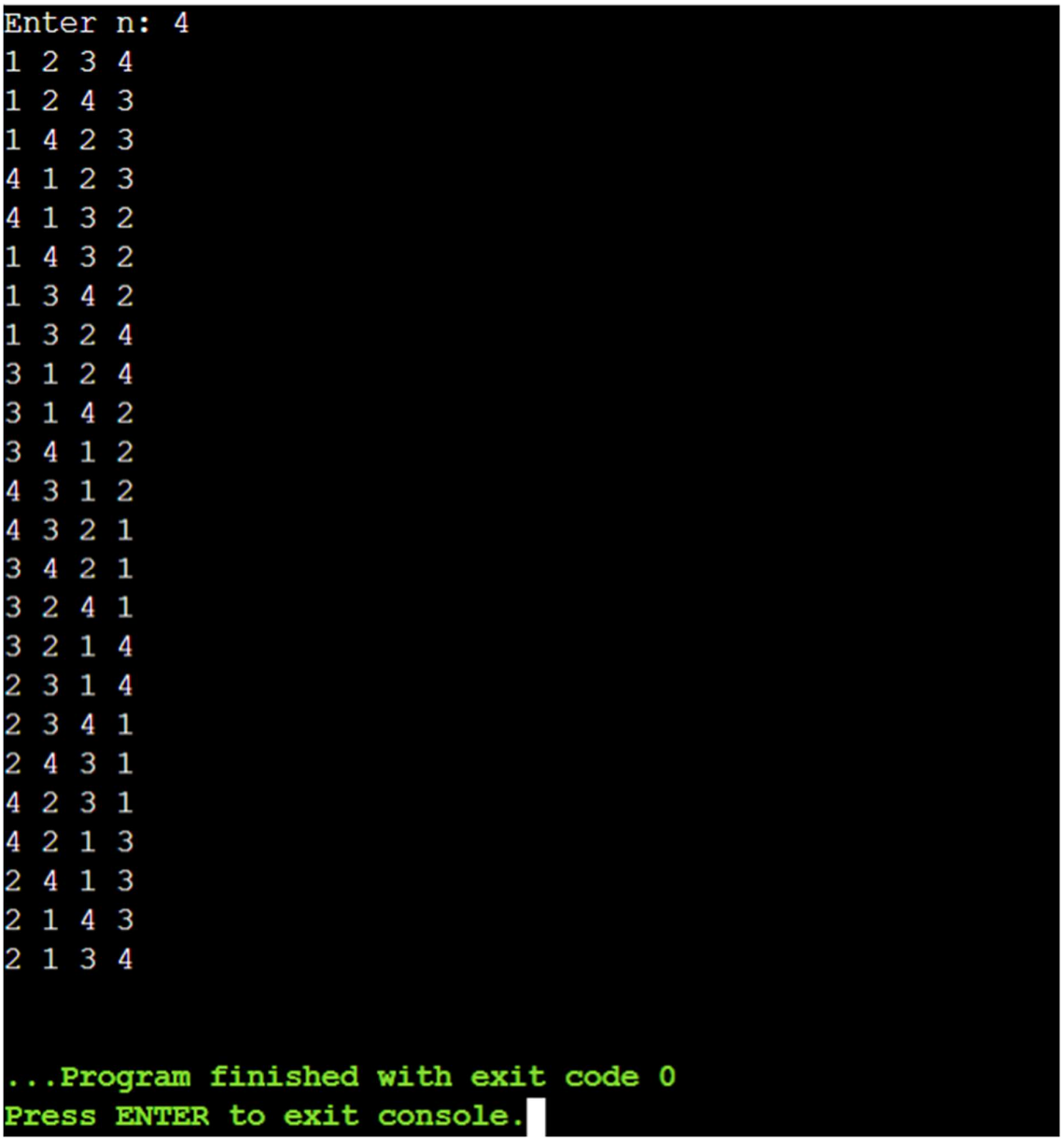
```
}
Enter number of vertices: 2
Enter adjacency matrix:
0 1
1 0
Topological Order: 0 1

...Program finished with exit code 0
Press ENTER to exit console.
```

Lab program 2:

Implement Johnson Trotter algorithm to generate permutations.

```
#include <stdio.h>

int a[10], d[10], n;

int mobile()
{
   int i, m = 0, p = -1;

   for (i = 0; i < n; i++)
   {
      if (d[i] == -1 && i - 1 >= 0 && a[i] > a[i - 1])
      {
         if (a[i] > m)
         {
            m = a[i];
            p = i;
         }
      }
      else if (d[i] == 1 && i + 1 < n && a[i] > a[i + 1])
      {
         if (a[i] > m)
         {
            m = a[i];
            p = i;
         }
      }
   }

   return p;
}

int main()
{
   int i, p, t, v;

   printf("Enter n: ");
   scanf("%d", &n);

   for (i = 0; i < n; i++)
   {
      a[i] = i + 1;
      d[i] = -1;
   }

   while (1)
   {
      for (i = 0; i < n; i++)
         printf("%d ", a[i]);

      printf("\n");

      p = mobile();

      if (p == -1)
         break;
```

```
        v = a[p];
        t = p + d[p];

        int x = a[p];
        a[p] = a[t];
        a[t] = x;

        x = d[p];
        d[p] = d[t];
        d[t] = x;

        for (i = 0; i < n; i++)
        {
            if (a[i] > v)
                d[i] = -d[i];
        }
    }

    return 0;
}
```


Enter n: 4
1 2 3 4
1 2 4 3
1 4 2 3
4 1 2 3
4 1 3 2
1 4 3 2
1 3 4 2
1 3 2 4
3 1 2 4
3 1 4 2
3 4 1 2
4 3 1 2
4 3 2 1
3 4 2 1
3 2 4 1
3 2 1 4
2 3 1 4
2 3 4 1
2 4 3 1
4 2 3 1
4 2 1 3
2 4 1 3
2 1 4 3
2 1 3 4
...Program finished with exit code 0
Press ENTER to exit console.

Lab program 3:

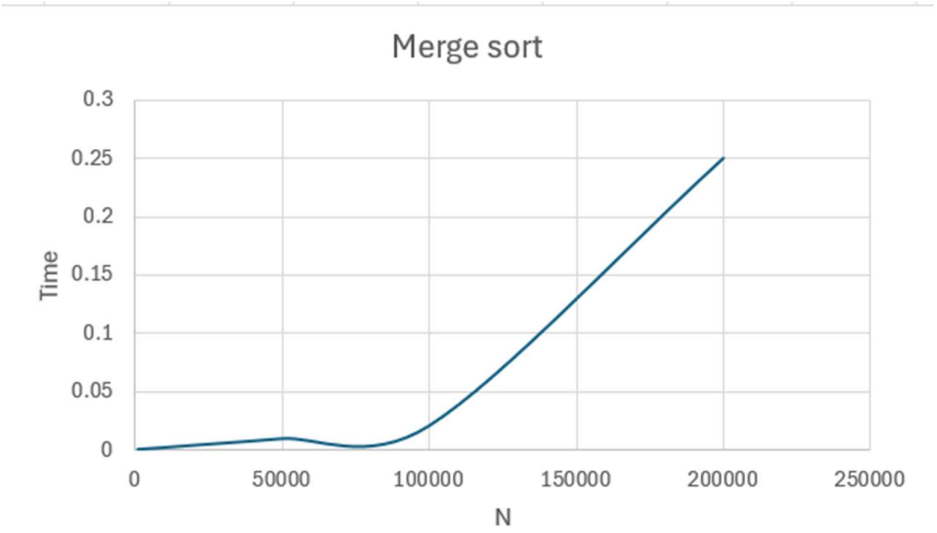
Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and record the time taken to sort.

```
#include<stdio.h>
#include<time.h>
void merge(int* a,int l,int m,int r){
   int c = l, b = m+1, k = 0;
   int v[50];
   while(c <= m && b <= r){
      if(a[c] < a[b]) v[k++] = a[c++];
      else v[k++] = a[b++];
   }
   while(c <= m)   v[k++] = a[c++];
   while(b <= r)     v[k++] = a[b++];
   for(int i=l,j=0;i<=r;i++,j++)  a[i] = v[j];
}
void mergesort(int l,int r,int* a){
   if(l >= r) return;
   int m = (l+r)/2;
   mergesort(l,m,a);
   mergesort(m+1,r,a);
   merge(a,l,m,r);
}
int main(){
   int n;
   int a[100];
   printf("Enter number of elements:");
   scanf("%d",&n);
   for (int i = 0; i < n; i++) {
      scanf("%d",&a[i]);
   }
   clock_t start,end;
   start = clock();
   mergesort(0,n-1,a);
   end = clock();     // end time
    printf("Sorted array:\n");
```

```
    for(int i=0;i<n;i++)  printf("%d ",a[i]);
    double time_taken = ((double)(end-start))/CLOCKS_PER_SEC;
    printf("\nTime taken: %f seconds\n",time_taken);
    return 0;
}
```

OUTPUT

```
Enter number of elements:9
3 4 1 3 5 6 2 2 1
Sorted array:
1 1 2 2 3 3 4 5 6
Time taken: 0.000017 seconds


...Program finished with exit code 0
Press ENTER to exit console.
```



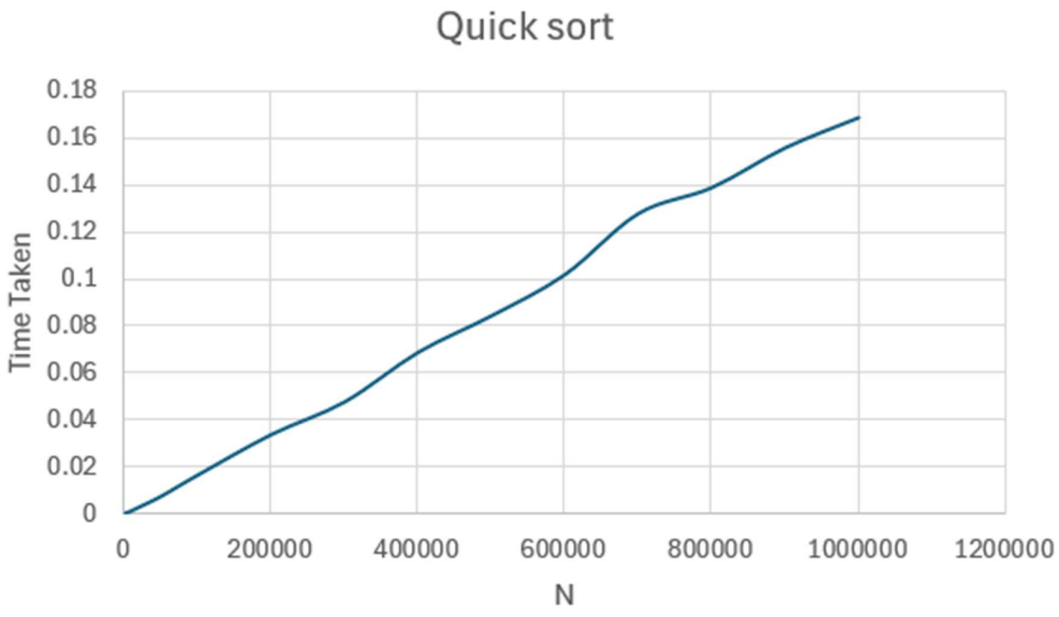
Lab program 4:

Sort a given set of N integer elements using Quick Sort technique and compute its time taken.

```
#include <stdio.h>
#include<time.h>
int partition(int* a, int l,int n){
  int i = l+1,j = n;
  while(1){
    while(i <= n && a[i] <= a[l]) i++;
    while(a[j] > a[l]) j--;
    if(i >= j) break;
    int t = a[i];
    a[i] = a[j];
    a[j] = t;
  }
  int t = a[l];
  a[l] = a[j];
  a[j] = t;
  return j;
}
void quicksort(int* a,int l,int n){
  if(l >= n) return;
  int pivotIdx = partition(a,l,n);
  quicksort(a,0,pivotIdx-1);
  quicksort(a,pivotIdx+1,n);
}
int main()
{
  int n;
  int a[100];
  printf("Enter number of elements:");
  scanf("%d",&n);
  for (int i = 0; i < n; i++) {
    scanf("%d",&a[i]);
  }
  clock_t start,end;
    start = clock();
```

```
  quicksort(a,0,n-1);
  end = clock();

 for (int i = 0; i < n; i++) {
  prin tf("%d ",a[i]);
 }
 double time_taken = (((double)(end-start))/CLOCKS_PER_SEC)*1000;
 printf("\nTime taken: %f milli seconds\n",time_taken);
}
```

```
Enter number of elements:9
23 343 2 1 3 4 4 5 1
1 1 2 3 4 4 5 23 343
Time taken: 0.002000 milli seconds


...Program finished with exit code 0
Press ENTER to exit console.
```



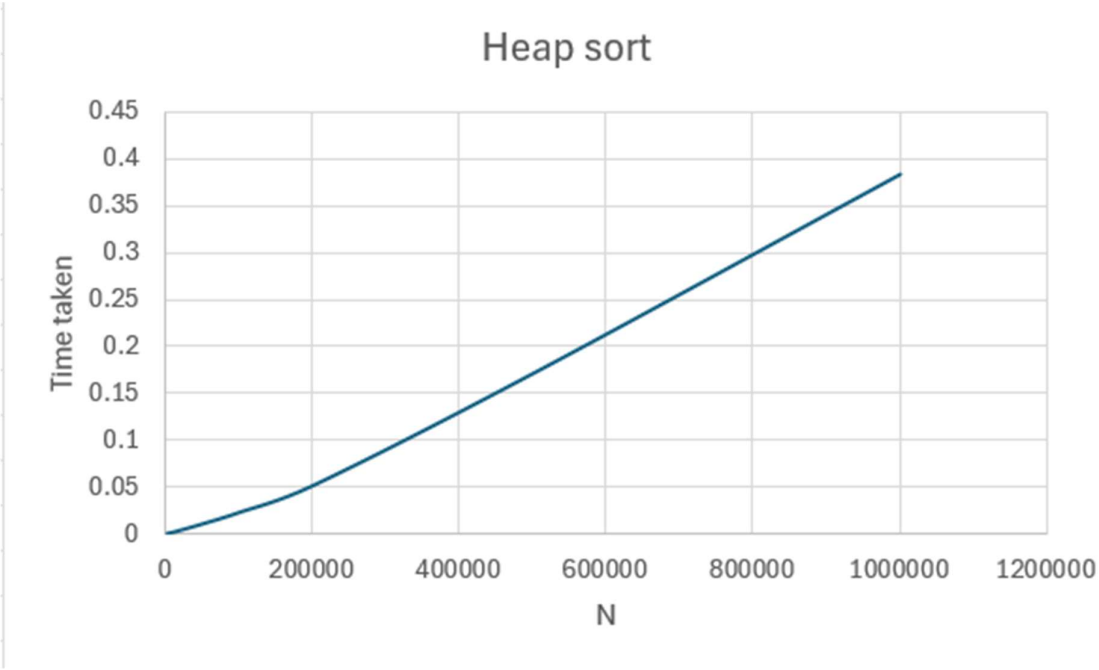
Lab program 5:

Sort a given set of N integer elements using Heap Sort technique and compute its time taken.

```
#include<stdio.h>
#include<time.h>

void heapify(int* a,int n,int p){
   int c,item;
   c = 2*p+1;
   item = a[p];
   while(c <= n-1){
            if(c+1 <= n-1){
         if(a[c] < a[c+1]) c++;
      }
      if(item < a[c]){
         a[p] = a[c];
         p = c;
         c = 2*p+1;
      }else break;
   }
   a[p] = item;
}
void buildheap(int* a, int n){
   for(int i = (n-1)/2; i >= 0; i--){
      heapify(a,n,i);
   }
}
void heapsort(int* a,int n){
   buildheap(a,n);
   for(int i = n-1; i > 0; i--){
      int t = a[i];
      a[i] = a[0];
      a[0] = t;
      heapify(a,i,0);
   }
}
int main(){
   int n;
   int a[100];
   printf("Enter number of elements:");
   scanf("%d",&n);
   for (int i = 0; i < n; i++) {
      scanf("%d",&a[i]);
   }
   clock_t start,end;
   start = clock();
   heapsort(a,n);
   end = clock();
   printf("Sorted array:\n");
   for(int i=0;i<n;i++)
      printf("%d ",a[i]);
```

```
    double time_taken = ((double)(end-start))/CLOCKS_PER_SEC;
    printf("\nTime taken: %f seconds\n",time_taken);
    return 0;
}
```

```
Enter number of elements:9
3 2 1 3 5 6 7 8 0
Sorted array:
0 1 2 3 3 5 6 7 8
Time taken: 0.000003 seconds


...Program finished with exit code 0
Press ENTER to exit console.
```



Lab program 6:

Implement 0/1 Knapsack problem using dynamic programming.

```
#include <stdio.h>

int max(int a, int b) {
   return (a > b) ? a : b;
}
int knapsack(int W, int wt[], int val[], int n) {
   int i, w;
   int K[n + 1][W + 1];

   for (i = 0; i <= n; i++) {
      for (w = 0; w <= W; w++) {
         if (i == 0 || w == 0)
            K[i][w] = 0;
         else if (wt[i - 1] <= w)
            K[i][w] = max(val[i - 1] + K[i - 1][w - wt[i - 1]], K[i - 1][w]);
         else
            K[i][w] = K[i - 1][w];
      }
   }
   return K[n][W];
}

int main() {
   int val[100], wt[100], n, W;

   printf("Enter number of items: ");
   scanf("%d", &n);

   printf("Enter values of items: ");
   for(int i = 0; i < n; i++) scanf("%d", &val[i]);
```

```
    printf("Enter weights of items: ");
    for(int i = 0; i < n; i++) scanf("%d", &wt[i]);

    printf("Enter capacity of knapsack: ");
    scanf("%d", &W);

    printf("Maximum profit: %d\n", knapsack(W, wt, val, n));
    return 0;
}
```

```
Enter number of items: 3
Enter values of items: 12 33 22
Enter weights of items: 2 3 4
Enter capacity of knapsack: 5
Maximum profit: 45


...Program finished with exit code 0
Press ENTER to exit console.
```

Lab program 7:

Implement All Pair Shortest paths problem using Floyd's algorithm.

```
#include<stdio.h>
#include<time.h>
int min(int a,int b){
   return a < b ? a : b;
}
void floyd(int a[][100],int n){
   for(int i=0;i<n;i++){
      for(int j=0;j<n;j++){
         for(int k=0;k<n;k++){
            a[j][k] = min(a[j][k],a[j][i]+a[i][k]);
         }
      }
   }
}
int main(){
   int n;
   int a[100][100];
   printf("Enter n:");
   scanf("%d",&n);
   for (int i = 0; i < n; i++) {
      for (int j = 0; j < n; j++) {
         scanf("%d",&a[i][j]);
      }
   }
   clock_t start,end;
   start = clock();
   floyd(a,n);
   end = clock();
   for(int i=0;i<n;i++){
      for(int j=0;j<n;j++)printf("%d ",a[i][j]);
```

```
        printf("\n");
    }

    double time_taken = ((double)(end-start))/CLOCKS_PER_SEC;

    printf("\nTime taken: %f seconds\n",time_taken);

    return 0;
}
```

```
Enter n:4
0 999999 6 1
4 0 10 5
999999 3 0 12
6 999999 12 0
0 9 6 1
4 0 10 5
7 3 0 8
6 15 12 0

Time taken: 0.000002 seconds


...Program finished with exit code 0
Press ENTER to exit console.
```

Lab program 8:

(a) Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.

```
#include<stdio.h>
#include<time.h>
struct edge {
   int s, e, cost;
};
int n, cost = 0, nv;
int vis[100];
struct edge a[100];
int main() {
   if (scanf("%d %d", &n, &nv) != 2) return 0;
   printf("Enter s, e and cost for following:\n");
   for (int i = 0; i < n; i++) {
      scanf("%d %d %d", &a[i].s, &a[i].e, &a[i].cost);
   }
   clock_t start,end;

   start = clock();
   vis[a[0].s] = 1;
   int edges_count = 0;

   while (edges_count < nv - 1) {
      int min = 100000;
      int minIdx = -1;

      for (int i = 0; i < n; i++) {
         if ((vis[a[i].s] && !vis[a[i].e]) || (!vis[a[i].s] && vis[a[i].e])) {
            if (a[i].cost < min) {
               min = a[i].cost;
               minIdx = i;
            }
         }
```

```
        }

        if (minIdx != -1) {
            cost += min;
            vis[a[minIdx].s] = 1;
            vis[a[minIdx].e] = 1;
            edges_count++;
        } else {
            break;
        }
    }
    end = clock();
    printf("The total cost is: %d\n", cost);
    double time_taken = ((double)(end-start))/CLOCKS_PER_SEC;

    printf("\nTime taken: %f seconds\n",time_taken);
    return 0;
}
```

```
9
10
Enter s, e and cost for following:
1 2 3
2 3 7
2 5 6
5 6 5
6 4 1
6 9 11
8 9 10
10 9 4
7 10 2
The total cost is: 49

Time taken: 0.000003 seconds
```

(b) Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.

```
#include<stdio.h>
#include<time.h>
struct edge{
   int s,e,cost;
};
int cost = 0;
 int grp[100];
 int find(int x){
   if(x != grp[x]) grp[x] = find(grp[x]);
   return grp[x];
 }
 void merge(int x,int y,int c){
   int gx = find(x);
   int gy = find(y);
   if(gx != gy){
      grp[gy] = gx;
      cost += c;
   }
 }
int main(){
   struct edge a[100];
   int n;
   scanf("%d",&n);
   printf("Enter s,e and cost for following");
   for(int i = 0; i < n; i++){
      scanf("%d",&a[i].s);
      scanf("%d",&a[i].e);
      scanf("%d",&a[i].cost);
   }
   clock_t start,end;

   start = clock();
```

```
    for(int i = 0; i < n; i++){
       for(int j = i+1; j < n; j++){
          if(a[i].cost > a[j].cost){
             struct edge t = a[i];
             a[i] = a[j];
             a[j] = t;
          }
       }
    }
    for(int i = 0; i < 100; i++) grp[i]=i;
    for(int i = 0; i < n; i++){
       merge(a[i].s,a[i].e,a[i].cost);
    }
    end = clock();
    printf("The total cost is: %d",cost);

    double time_taken = ((double)(end-start))/CLOCKS_PER_SEC;

    printf("\nTime taken: %f seconds\n",time_taken);
    return 0;
}
```

```
9
Enter s,e and cost for following1 2 3
2 3 7
2 5 6
5 6 5
6 4 1
6 9 11
8 9 10
10 9 4
7 10 2
The total cost is: 49
Time taken: 0.000004 seconds


...Program finished with exit code 0
Press ENTER to exit console.
```

Lab program 9:
Implement Fractional Knapsack using Greedy technique.

```
#include <stdio.h>

void swap(float *a, float *b)
{
   float t = *a;
   *a = *b;
   *b = t;
}

int main()
{
   int n;

   printf("Enter no of elements, prices and weights:\n");
   scanf("%d", &n);

   float p[100], wt[100], temp[100];

   for (int i = 0; i < n; i++)
      scanf("%f", &p[i]);

   for (int i = 0; i < n; i++)
   {
      scanf("%f", &wt[i]);

      if (wt[i] == 0)
         return 0;

      temp[i] = p[i] / wt[i];
   }

   /* Sort in descending order of profit/weight ratio */
   for (int i = 0; i < n; i++)
   {
      for (int j = i + 1; j < n; j++)
      {
         if (temp[i] < temp[j])
         {
            swap(&p[i], &p[j]);
            swap(&temp[i], &temp[j]);
            swap(&wt[i], &wt[j]);
         }
      }
   }

   int m;
   float res = 0;

   printf("Enter capacity: ");
   scanf("%d", &m);

   for (int i = 0; i < n; i++)
```

```
    {
        if (m <= 0)
            break;

        float x;

        if (m < wt[i])
            x = m;
        else
            x = wt[i];

        res += temp[i] * x;
        m -= x;
    }

    printf("The maximum profit is %.2f\n", res);

    return 0;
}
```

```
Enter no of elements, prices and weights:
3
1 2 1
2 1 2
Enter capacity: 2
The maximum profit is 2.50


...Program finished with exit code 0
Press ENTER to exit console.
```

Lab program 10:

From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.

```
#include<stdio.h>
void swap(float* a,float* b){
   float t = *a;
   *a = *b;
   *b = t;
}
void main(){
   int n;
   printf("Enter no of vertices");
   scanf("%d",&n);
   int dist[100], adj[2][100][100],vis[100];
   for(int i = 0; i < n; i++){
      dist[i] = 10000;
      vis[i] = 0;
      for(int j = 0; j < n; j++){
         scanf("%d",&adj[0][i][j]);
         scanf("%d",&adj[1][i][j]);
      }
   }
   dist[0] = 0;
   for(int s = 0; s < n; s++){
      int curr = 10001,idx = 0;
      for(int j = 0; j < n; j++){
         if(!vis[j] && curr > dist[j]){
            curr = dist[j];
            idx = j;
         }
      }
      for(int i = 0; i < n; i++){
         if(!vis[i] && adj[0][idx][i] && dist[i] > dist[idx]+adj[1][idx][i]){
            dist[i] = dist[idx]+adj[1][idx][i];
         }
      }
      vis[idx] = 1;
   }
   for(int i = 0; i < n; i++){
      printf("The min distance is for %d: %d \n",i,dist[i]);
   }
}
```

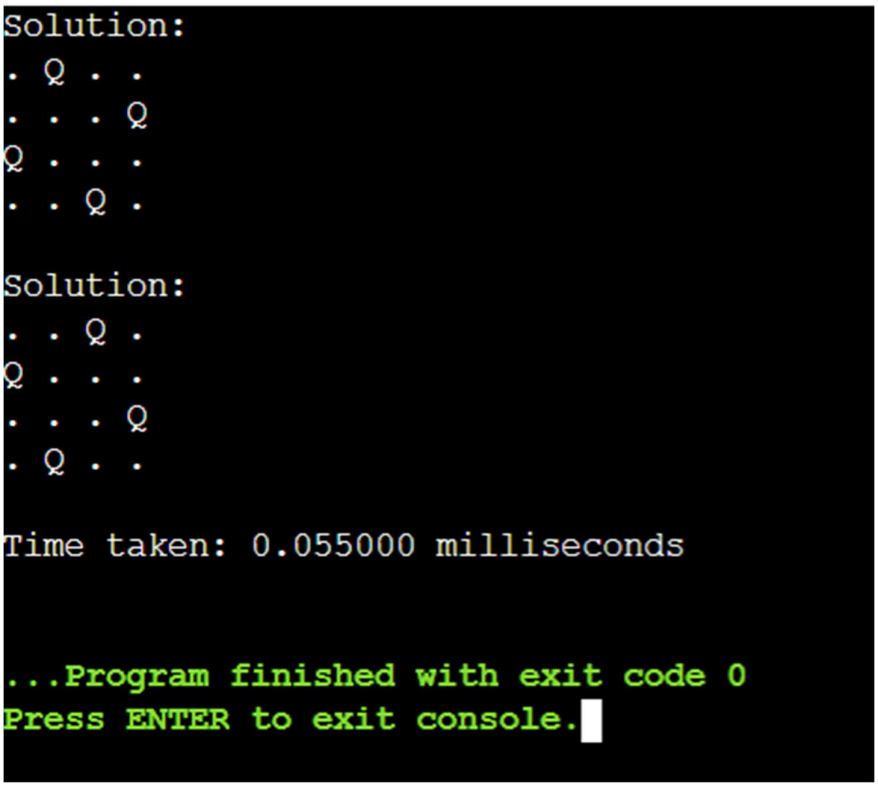
```
Enter no of vertices4
0 0 1 1 0 1 2 0
0 0 0 0 0 0 0 0
2 1 0 0 1 2 0 0
0 0 1 2 0 0 0 0
The min distance is for 0: 0
The min distance is for 1: 1
The min distance is for 2: 10000
The min distance is for 3: 0


...Program finished with exit code 4
Press ENTER to exit console.
```

Lab program 10:

Implement “N-Queens Problem” using Backtracking.

```
#include <stdio.h>
#include <time.h>
#include <math.h>

int mat[100];

int checkValid(int row, int col) {
   for(int i = 0; i < row; i++) {
      if(mat[i] == col || abs(i - row) == abs(mat[i] - col))
         return 0;
   }
   return 1;
}

void nqueens(int row, int n) {
   if(row == n) {
      printf("Solution:\n");

      for(int i = 0; i < n; i++) {
         for(int j = 0; j < n; j++) {
            if(mat[i] == j)
               printf("Q ");
            else
               printf(". ");
         }
         printf("\n");
      }

      printf("\n");
      return;
   }

   for(int col = 0; col < n; col++) {
      if(checkValid(row, col)) {
         mat[row] = col;
         nqueens(row + 1, n);
      }
   }
}

int main() {
   int n = 4;
```

```
    clock_t start, end;
    start = clock();

    nqueens(0, n);

    end = clock();

    double time_taken =
       ((double)(end - start) / CLOCKS_PER_SEC) * 1000;

    printf("Time taken: %f milliseconds\n", time_taken);

    return 0;
}
```



11. Container With Most Water

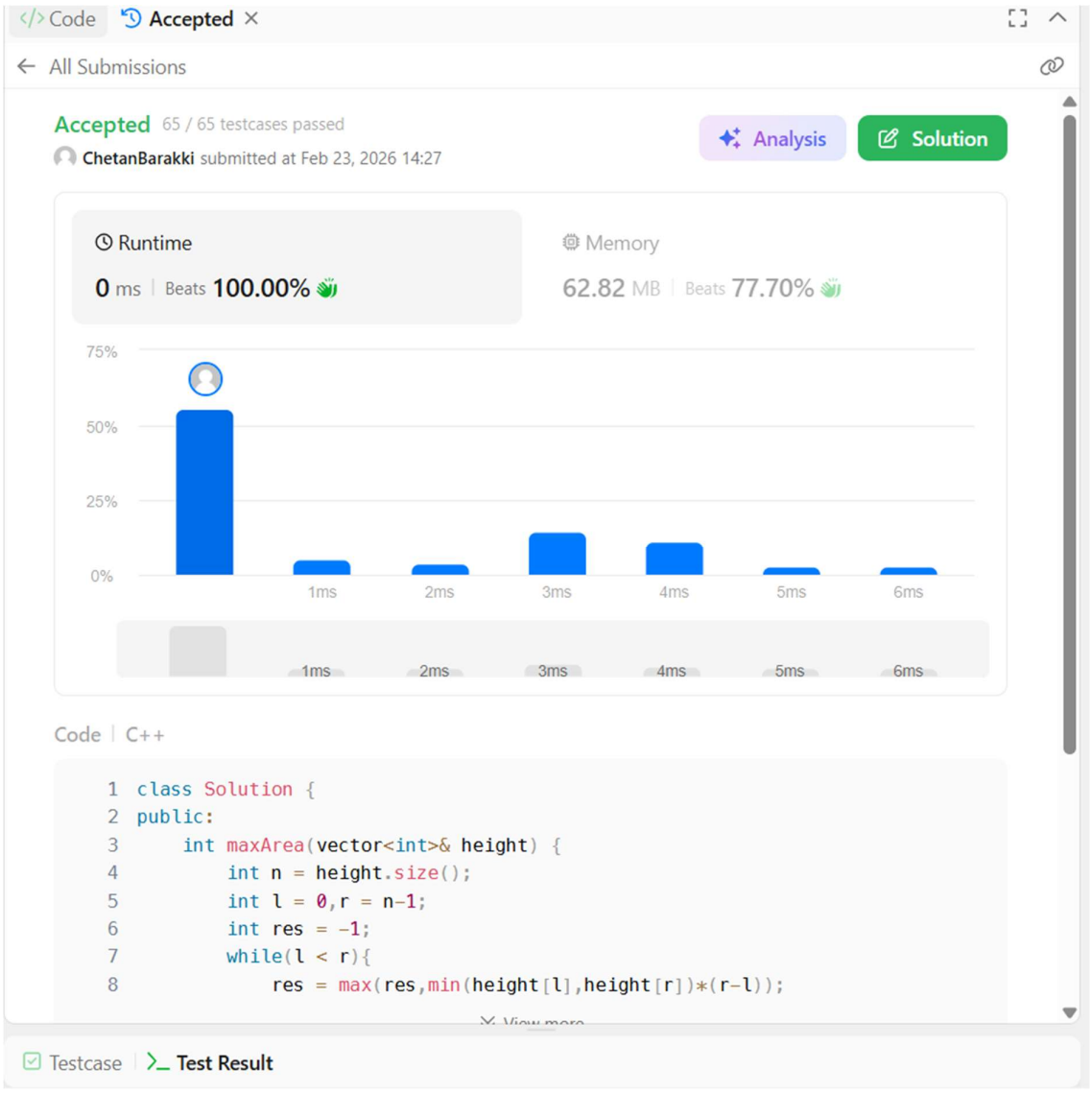
You are given an integer array height of length n. There are n vertical lines drawn such that the two endpoints of the i^{th} line are (i, 0) and (i, height[i]).

Find two lines that together with the x-axis form a container, such that the container contains the most water.

Return *the maximum amount of water a container can store*.

```
#include <stdio.h>
static inline int max_int(int a, int b) {
    return (a > b) ? a : b;
}
static inline int min_int(int a, int b) {
    return (a < b) ? a : b;
}
int maxArea(int* height, int heightSize) {
    int l = 0;
    int r = heightSize - 1;
    int res = -1;
    while (l < r) {
        int current_height = min_int(height[l], height[r]);
        int current_width = r - l;
        int area = current_height * current_width;
        res = max_int(res, area);
        if (height[l] < height[r]) {
            l++;
        } else {
            r--;
        }
    }
    return res;
```

}


Code
Accepted
All Submissions
Accepted 65 / 65 testcases passed
ChetanBarakki submitted at Feb 23, 2026 14:27
Analysis
Solution
Runtime
0 ms | Beats 100.00%
Memory
62.82 MB | Beats 77.70%
75%
50%
25%
0%
1ms
2ms
3ms
4ms
5ms
6ms
Code | C++
1 class Solution {
2 public:
3 int maxArea(vector<int>& height) {
4 int n = height.size();
5 int l = 0,r = n-1;
6 int res = -1;
7 while(l < r){
8 res = max(res,min(height[l],height[r])*(r-l));
Testcase
Test Result

34. Find First and Last Position of Element in Sorted Array

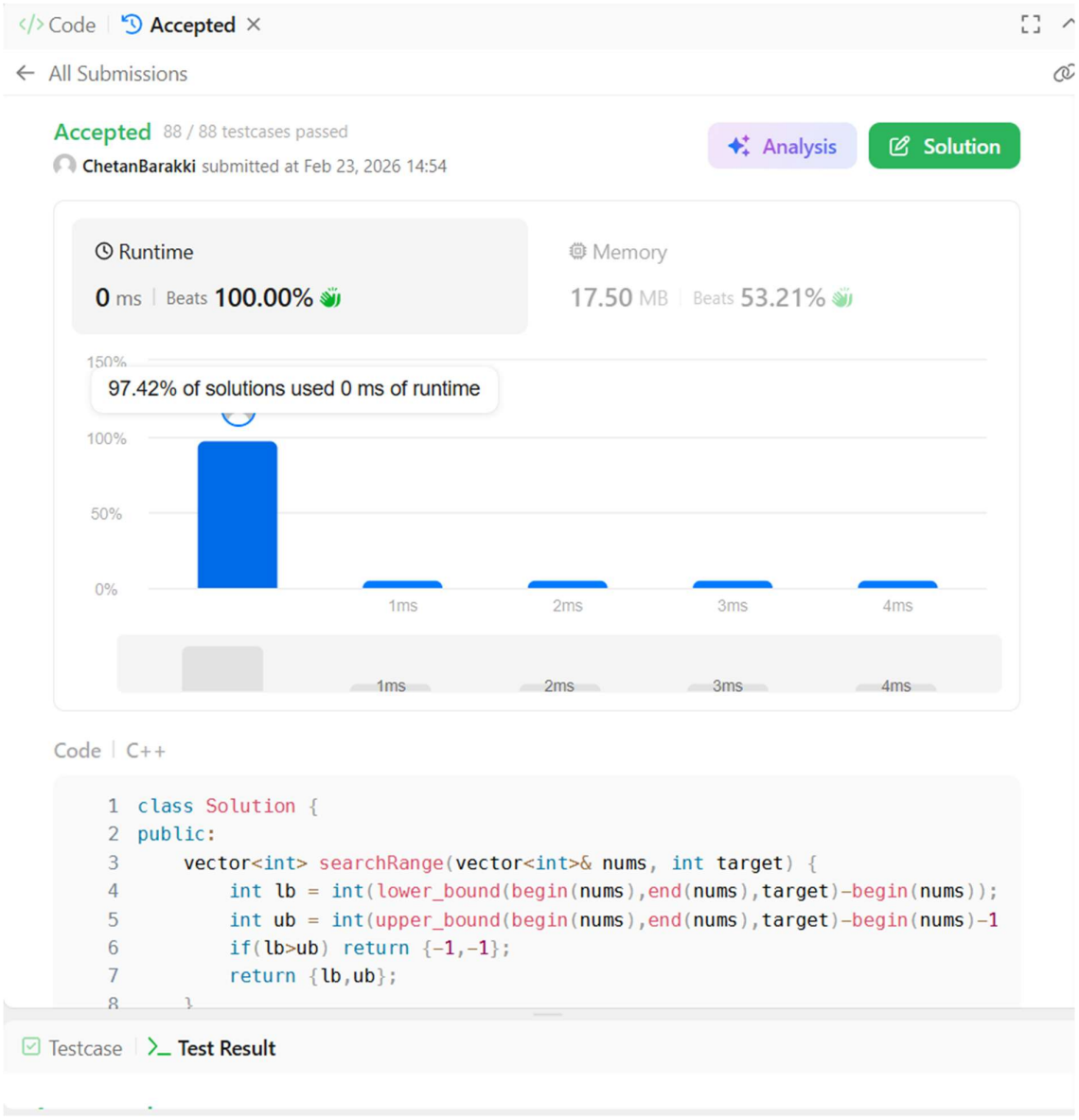
Given an array of integers nums sorted in non-decreasing order, find the starting and ending position of a given target value.

If target is not found in the array, return [-1, -1].

You must write an algorithm with O(log n) runtime complexity.

```
 #include <stdio.h>
#include <stdlib.h>
int* searchRange(int* nums, int numsSize, int target, int* returnSize) {
    *returnSize = 2;
    int* result = (int*)malloc(2 * sizeof(int));
    result[0] = -1;
    result[1] = -1
    if (numsSize == 0) return result;
    int l = 0, r = numsSize - 1, idx = -1;
    while (l <= r) {
        int mid = l + (r - l) / 2;
        if (nums[mid] == target) {
            idx = mid;
            break;
        } else if (nums[mid] > target) {
            r = mid - 1;
        } else {
            l = mid + 1;
        }
    }
    if (idx == -1) return result;
    int i = idx, j = idx;
    while (i + 1 < numsSize && nums[i + 1] == target) i++;
    while (j - 1 >= 0 && nums[j - 1] == target) j--;
```

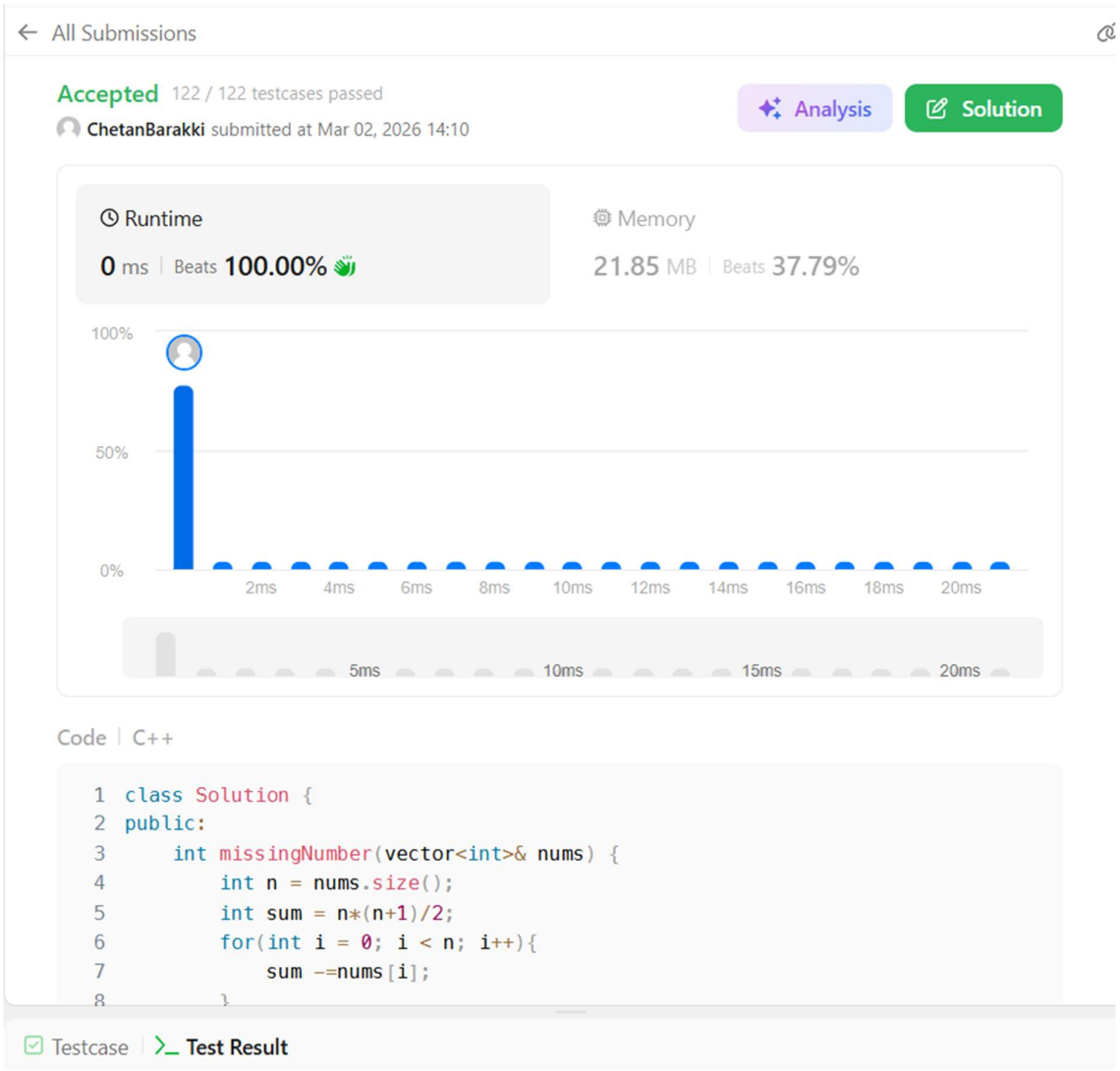
```
    result[0] = j;
    result[1] = i;
    return result;
}
```



268. Missing Number

Given an array nums containing n distinct numbers in the range [0, n], return *the only number in the range that is missing from the array.*

```
int missingNumber(int* nums, int numsSize) {
    int sum = numsSize * (numsSize + 1) / 2;
    for (int i = 0; i < numsSize; i++) {
        sum -= nums[i];
    }
    return sum;
}
```



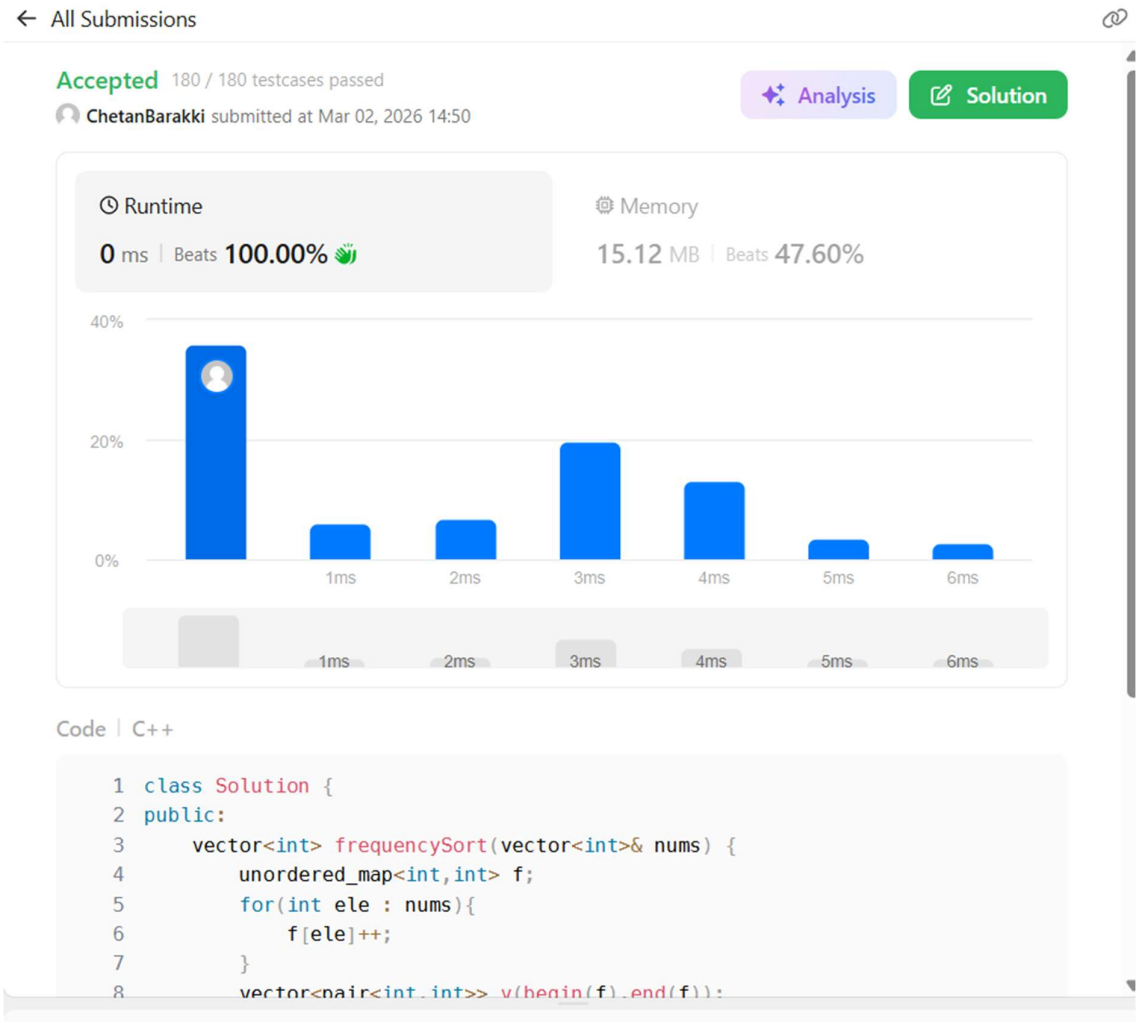
1636. Sort Array by Increasing Frequency

Given an array of integers nums, sort the array in increasing order based on the frequency of the values. If multiple values have the same frequency, sort them in decreasing order.

Return the *sorted array*.

```
#include <stdlib.h>

typedef struct {
    int val;
    int freq;
} Element;

int compare(const void* a, const void* b) {
    Element* e1 = (Element*)a;
    Element* e2 = (Element*)b;

    if (e1->freq != e2->freq) {
        return e1->freq - e2->freq;
    }
    return e2->val - e1->val;
}

int* frequencySort(int* nums, int numsSize, int* returnSize) {
    *returnSize = numsSize;
    if (numsSize == 0) return nums;

    int temp_arr[numsSize];
    for (int i = 0; i < numsSize; i++) temp_arr[i] = nums[i];

    // Simple sort to group identical elements
    for (int i = 0; i < numsSize - 1; i++) {
        for (int j = 0; j < numsSize - i - 1; j++) {
            if (temp_arr[j] > temp_arr[j + 1]) {
                int temp = temp_arr[j];
                temp_arr[j] = temp_arr[j + 1];
                temp_arr[j + 1] = temp;
            }
        }
    }
    Element elements[numsSize];
    int count = 0;
    for (int i = 0; i < numsSize; ) {
        int j = i;
        while (j < numsSize && temp_arr[j] == temp_arr[i]) j++;
        elements[count].val = temp_arr[i];
        elements[count].freq = j - i;
        count++;
        i = j;
    }

    qsort(elements, count, sizeof(Element), compare);
```

```
    int k = 0;
    for (int i = 0; i < count; i++) {
        for (int f = 0; f < elements[i].freq; f++) {
            nums[k++] = elements[i].val;
        }
    }

    return nums;
}
```



207. Course Schedule

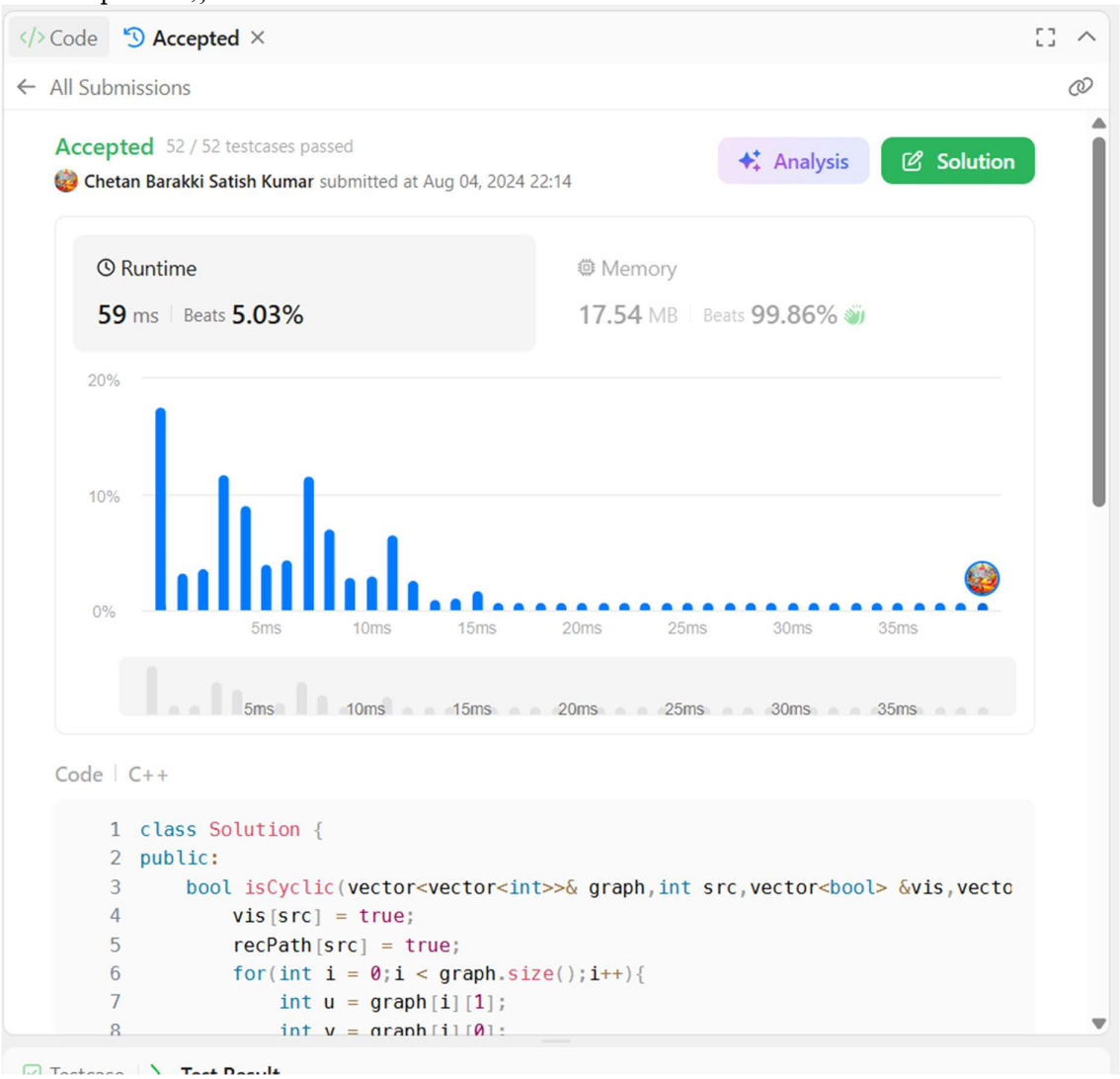
There are a total of numCourses courses you have to take, labeled from 0 to numCourses - 1. You are given an array prerequisites where prerequisites[i] = [a_i, b_i] indicates that you must take course b_i first if you want to take course a_i.

- **For example, the pair [0, 1], indicates that to take course 0 you have to first take course 1.**

Return true if you can finish all courses. Otherwise, return false.

```
#include <stdbool.h>
#include <stdlib.h>
// Simple adjacency list node
typedef struct Node {
    int dest;
    struct Node* next;
} Node;

bool isCyclic(int src, Node** adj, bool* vis, bool* recPath) {
    vis[src] = true;
    recPath[src] = true;
    Node* temp = adj[src];
    while (temp != NULL) {
        int v = temp->dest;
        if (!vis[v]) {
            if (isCyclic(v, adj, vis, recPath)) return true;
        } else if (recPath[v]) {
            return true;
        }
        temp = temp->next;
    }

    recPath[src] = false;
    return false;
}

bool canFinish(int numCourses, int** prerequisites, int prerequisitesSize, int* prerequisitesColSize) {
    // Build Adjacency List
    Node** adj = (Node**)calloc(numCourses, sizeof(Node*));
    for (int i = 0; i < prerequisitesSize; i++) {
        int u = prerequisites[i][1];
        int v = prerequisites[i][0];
        Node* newNode = (Node*)malloc(sizeof(Node));
        newNode->dest = v;
        newNode->next = adj[u];
        adj[u] = newNode;
    }

    bool* vis = (bool*)calloc(numCourses, sizeof(bool));
    bool* recPath = (bool*)calloc(numCourses, sizeof(bool));

    bool possible = true;
    for (int i = 0; i < numCourses; i++) {
        if (!vis[i]) {
```

```
        if (isCyclic(i, adj, vis, recPath)) {
            possible = false;
            break;
        }
    }
}

// Cleanup memory
for (int i = 0; i < numCourses; i++) {
    Node* curr = adj[i];
    while (curr) {
        Node* temp = curr;
        curr = curr->next;
        free(temp);
    }
}
free(adj);
free(vis);
free(recPath);

return possible;}
```



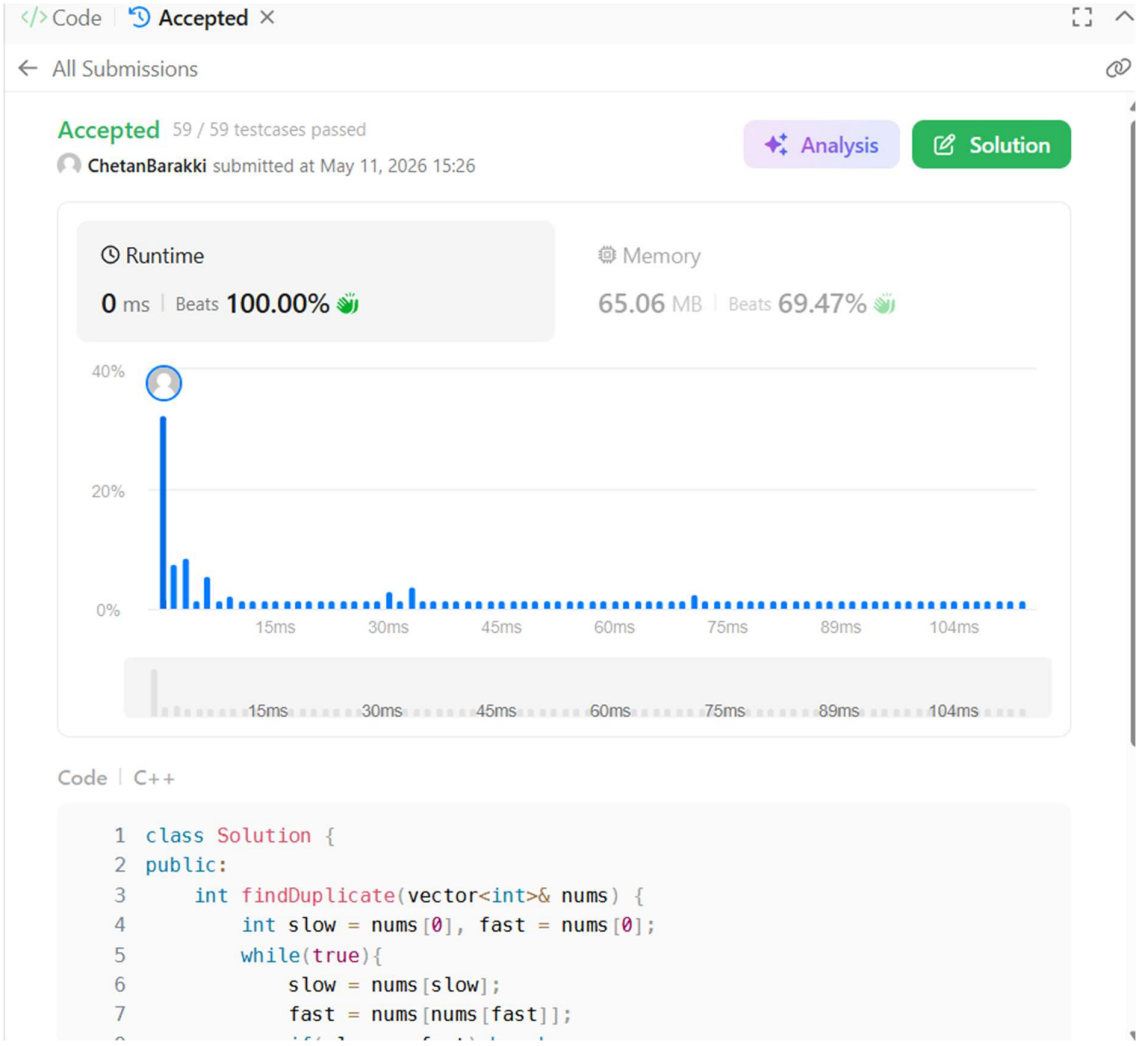
287. Find the Duplicate Number

Given an array of integers nums containing n + 1 integers where each integer is in the range [1, n] inclusive.

There is only one repeated number in nums, return *this repeated number*.

You must solve the problem without modifying the array nums and using only constant extra space.

```
int findDuplicate(int* nums, int numsSize) {
    int slow = nums[0];
    int fast = nums[0];

    while (1) {
        slow = nums[slow];
        fast = nums[nums[fast]];
        if (slow == fast) break;
    }

    slow = nums[0];
    while (slow != fast) {
        slow = nums[slow];
        fast = nums[fast];
    }

    return slow;
}
```


Code
Accepted
All Submissions
Accepted 59 / 59 testcases passed
ChetanBarakki submitted at May 11, 2026 15:26
Analysis
Solution
Runtime
0 ms | Beats 100.00%
Memory
65.06 MB | Beats 69.47%
40%
20%
0%
15ms
30ms
45ms
60ms
75ms
89ms
104ms
Code | C++
class Solution {
public:
int findDuplicate(vector<int>& nums) {
int slow = nums[0], fast = nums[0];
while(true){
slow = nums[slow];
fast = nums[nums[fast]];

801. Minimum Swaps To Make Sequences Increasing

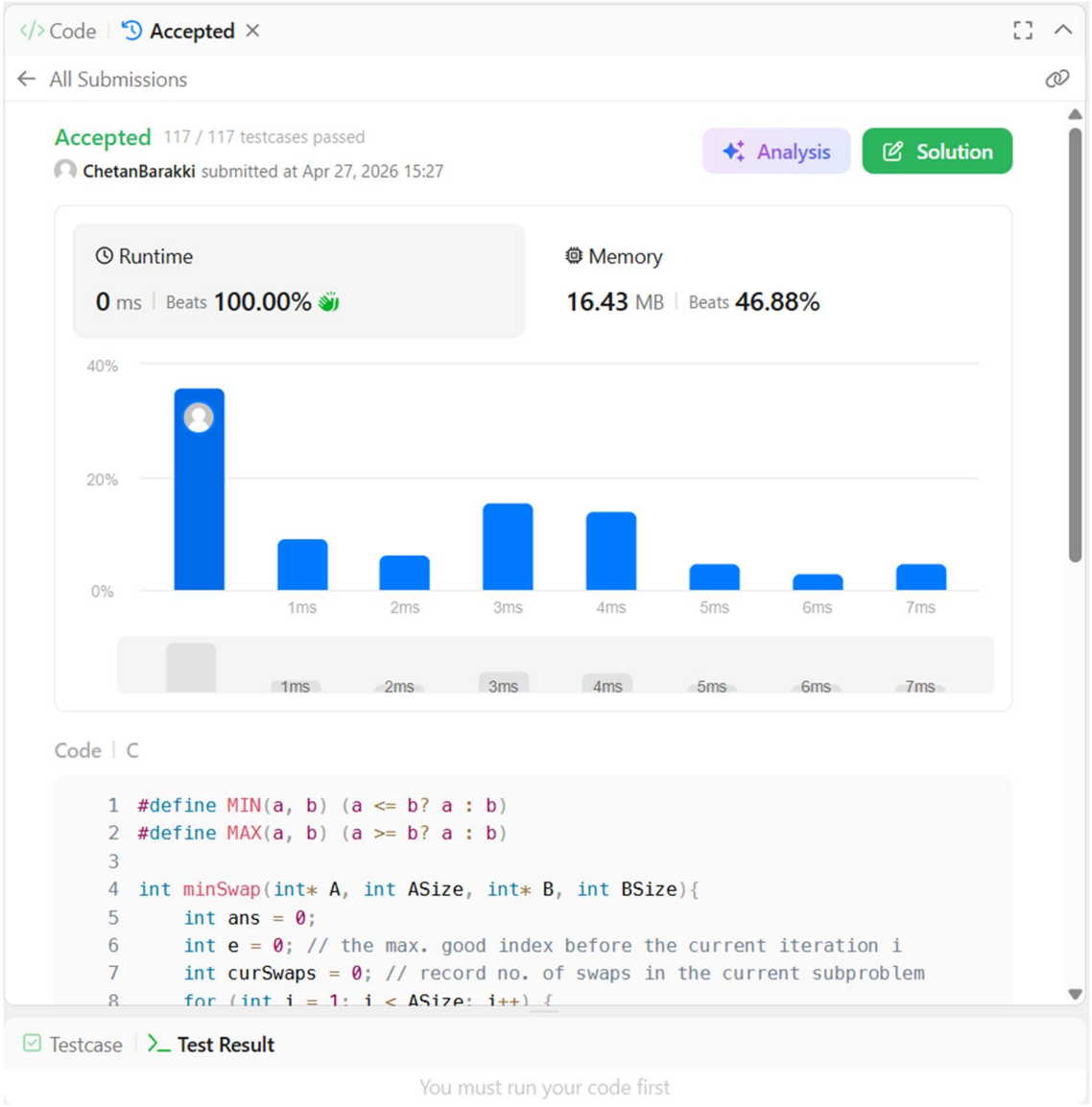
You are given two integer arrays of the same length nums1 and nums2. In one operation, you are allowed to swap nums1[i] with nums2[i].

- **For example, if nums1 = [1,2,3,8], and nums2 = [5,6,7,4], you can swap the element at i = 3 to obtain nums1 = [1,2,3,4] and nums2 = [5,6,7,8].**

Return *the minimum number of needed operations to make* nums1 *and* nums2 *strictly increasing*. The test cases are generated so that the given input always makes it possible.

An array arr is strictly increasing if and only if arr[0] < arr[1] < arr[2] < ... < arr[arr.length - 1].

```
#define MIN(a, b) (a <= b? a : b)
#define MAX(a, b) (a >= b? a : b)

int minSwap(int* A, int ASize, int* B, int BSize){
   int ans = 0;
   int e = 0; // the max. good index before the current iteration i
   int curSwaps = 0; // record no. of swaps in the current subproblem
   for (int i = 1; i < ASize; i++) {
      if (MIN(A[i], B[i]) > MAX(A[i-1], B[i-1])) { // if i is a good index
         ans += MIN(curSwaps, i-e-curSwaps); // select min. swaps of the two configurations
         e = i;
         curSwaps = 0;
         continue;
      }
      if (A[i-1] >= A[i] || B[i-1] >= B[i]) { // if need to swap i
         int temp = A[i];
         A[i] = B[i];
         B[i] = temp;
         curSwaps++;
      }
   } // end for
   ans += MIN(curSwaps, ASize-e-curSwaps);
   return ans;
}
```


Code
Accepted
All Submissions
Accepted 117 / 117 testcases passed
ChetanBarakki submitted at Apr 27, 2026 15:27
Analysis
Solution
Runtime
0 ms | Beats 100.00%
Memory
16.43 MB | Beats 46.88%
40%
20%
0%
1ms
2ms
3ms
4ms
5ms
6ms
7ms
Code | C
#define MIN(a, b) (a <= b? a : b)
#define MAX(a, b) (a >= b? a : b)
int minSwap(int* A, int ASize, int* B, int BSize){
int ans = 0;
int e = 0; // the max. good index before the current iteration i
int curSwaps = 0; // record no. of swaps in the current subproblem
Testcase | Test Result
You must run your code first